

216301013

BCE-P517

SEMESTER EXAMINATION-DECEMBER 2023

• CLASS: BTech SEMESTER: V •

MACHINE LEARNING-I

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: A and B. Attempt both the sections as per given instructions.

SECTION-A

(SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

- Question-1: What are the different types of Machine Learning? Also explain their use cases.
- Question-2: What are the types of problem? Explain the difference between Regression and Classification.
- Question-3: What are the different types of data? Explain the difference between Structured Data and Unstructured Data.
- Question-4: What are the different Popular Machine Learning Datasets? Explain.
- Question-5: What are the most commonly used Data repositories sources for machine learning practices? Explain each.
- Question-6: What is the Importance of cross validation? Explain Holdout Method and K-fold cross validation.
- Question-7: What do you mean by Fine tuning of model? Explain the difference between Grid Search.
- Question-8: What do you understand by Batch and online learning? Explain.
- Question-9: What are the applications of machine learning? Explain.
- Question-10: What is reinforcement learning? Explain its types.

SECTION-B
(LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.
(4 X 10 = 40 Marks)

- Question-1: What is Machine Learning? What is the difference between Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL)? Explain.
- Question-2: What are the different Python Libraries? Explain different frame work for machine learning algorithm.
- Question-3: Explain the Basic concept of Probability theory and Linear Algebra, Bias, Variance, Bias-Variance trade-off, overfitting and underfitting.
- Question-4: How will you perform Pre-processing of data like Data cleaning, wrangling and filtering, Handling missing and categorical data?
- Question-5: Explain a case study how ML can be used to predict data in stock merit.
- Question-6: What do you mean by the performance metrics for Machine Learning Algorithm? Explain Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Confusion Matrix with help of example.
- Question-7: What is regression problems? Explain different types of regression with help of suitable example.
- Question-8: What is classification? Explain the difference between Binary Classification, Multi-Class Classification, Multi-Label Classification, Imbalanced Classification.